

3-Mark Questions & Answers -- DBMS Module 2: Relational Algebra & Calculus

Q1. Explain Set Difference () and Intersection (∩) operations with examples.

1. Set Difference (R − S): Returns tuples that belong to relation R but NOT to relation S (both must be union-compatible).

2. Int

Q2. Explain Natural Join (JOIN) vs Theta Join (JOIN).

Feature	Natural Join (JOIN)	Theta Join (JOIN)
Join Condition	Equality on ALL common attribute names	Explicit conditional predicate (e.g.,
Duplicate Columns	Eliminates duplicate common columns in	Retains columns from both relations un

Q3. Differentiate between Procedural and Non-Procedural query languages.

Feature	Relational Algebra (Procedural)	Relational Calculus (Non-Procedural)
Approach	Tells **HOW** to retrieve data step-by	Tells **WHAT** data to retrieve withou
Execution	Specifies exact sequence of relational	Uses first-order predicate calculus lo
Foundation	Basis for DBMS physical query executio	Theoretical foundation for SQL declara